

Lab Assignment – 8

Neural Networks and Multilayer Perceptrons

Question 1: Effect of Hidden Layer Size in a Multilayer Perceptron

Objective: The aim of this experiment is to study how the number of neurons in the hidden layer affects the learning behaviour and performance of a Multilayer Perceptron (MLP) classifier.

Dataset generation: Create a synthetic binary classification dataset containing at least 800 samples. Generate input variables x_1 and x_2 from a uniform distribution $U(-2, 2)$. The intermediate variable is defined as:

$$z = \sin(1.5x_1) + x_2^2 - 0.5x_1x_2 + \varepsilon$$

where $\varepsilon \sim N(0, 0.2)$. The class label is assigned as $y = 1$ if $z > 0$, otherwise $y = 0$.

Common experimental settings: Use a train–test split of 70% and 30%, gradient descent optimization, 120 training epochs, and learning rate 0.01. Use ReLU as the hidden layer activation function and Sigmoid as the output activation function. Record training loss, testing loss, training accuracy, and testing accuracy at every epoch.

Task: Construct MLP models with a single hidden layer. Train separate models where the number of hidden neurons $h \in \{2, 4, 8, 16, 32, 64\}$. For each configuration, train the model for 120 epochs and record the performance metrics at every epoch.

Plotting: For each value of h , create an epoch-wise plot where the x-axis represents the training epoch and the y-axis represents loss. Include both training loss and testing loss curves. After training completes, create another plot showing the relationship between the number of hidden neurons and the final accuracy.

Question 2: Effect of Network Depth in Multilayer Perceptrons

Objective: This experiment investigates how increasing the number of hidden layers (network depth) affects the learning capacity and generalization performance of an MLP.

Dataset: Use the dataset generated in Question 1.

Experimental settings: Use a 70%–30% train–test split, gradient descent optimization, 120 training epochs, and learning rate 0.01. Use ReLU activation for hidden layers and Sigmoid activation for the output layer. Each hidden layer should contain 16 neurons.

Model	Architecture
M1	$2 \rightarrow 16 \rightarrow 1$
M2	$2 \rightarrow 16 \rightarrow 16 \rightarrow 1$
M3	$2 \rightarrow 16 \rightarrow 16 \rightarrow 16 \rightarrow 1$
M4	$2 \rightarrow 16 \rightarrow 16 \rightarrow 16 \rightarrow 16 \rightarrow 1$

For each model, train the network for 120 epochs and record training loss, testing loss, training accuracy and testing accuracy at each epoch.

Question 3: Decision Boundary Learning in Multilayer Perceptrons

Dataset generation: Generate 600 samples where $x_1, x_2 \sim U(-3, 3)$. The intermediate variable is defined as:

$$z = x_1^2 + x_2^2 - 4 + \varepsilon$$

where $\varepsilon \sim N(0, 0.5)$. Assign class label $y = 1$ if $z > 0$, otherwise $y = 0$.

Split the dataset into 70% training data and 30% testing data and train an MLP with architecture $2 \rightarrow 16 \rightarrow 1$ using ReLU activation in the hidden layer and Sigmoid activation in the output layer.

Question 4: XOR Classification using a Multilayer Perceptron

Implement a multilayer perceptron with architecture $2 \rightarrow 6 \rightarrow 1$ to solve the XOR classification problem using the backpropagation algorithm. Include a bias term at every neuron. Use ReLU activation in the hidden layer and Sigmoid activation in the output layer.

Train the model for 100 epochs and record the training loss at each epoch. Finally, plot the convergence graph showing epoch versus training loss.